

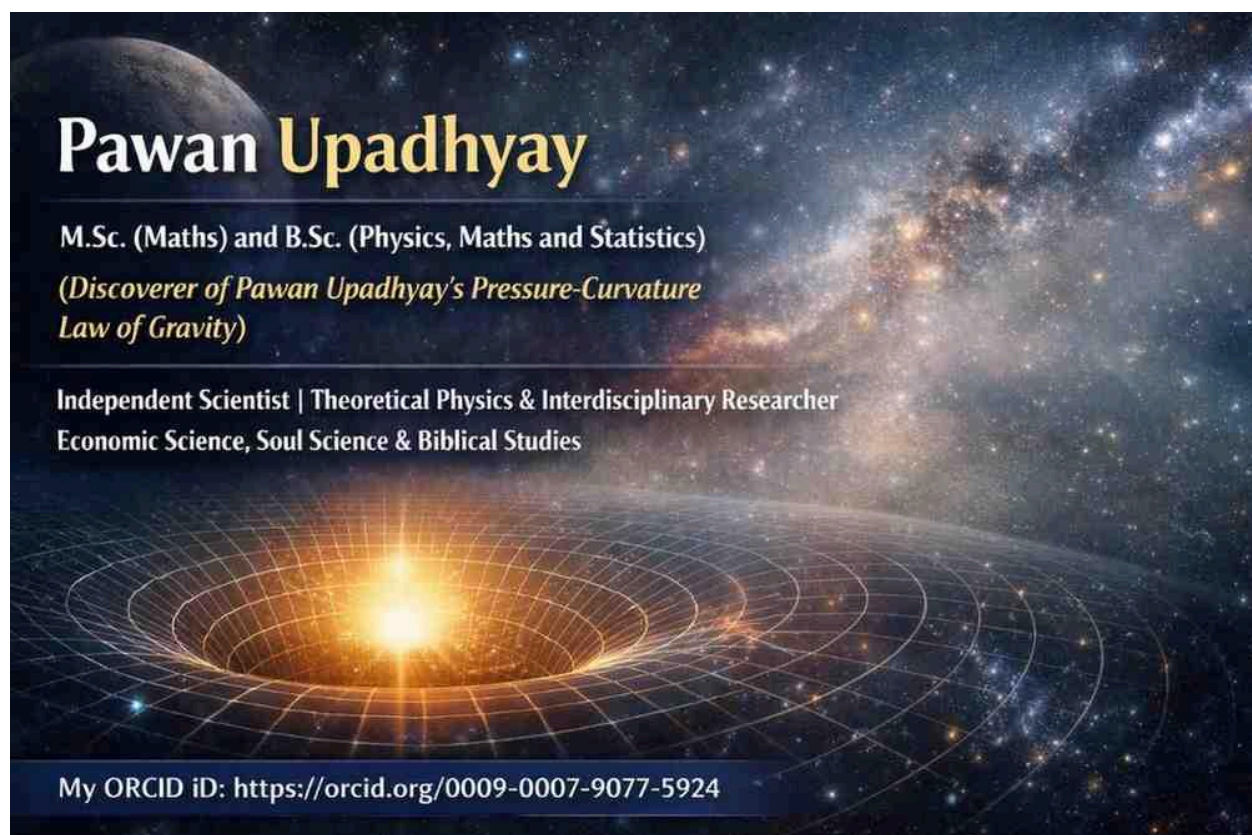
Gravitational Pressure Field, Pressure Gradient Field, and Gravitational Field: A Unified PPC Framework

Author: Pawan Upadhyay

Affiliation: Independent Researcher

Email: pawanupadhyay28@hotmail.com

ORCID iD: <https://orcid.org/0009-0007-9077-5924>



Abstract

This paper presents a unified formulation of gravity within the Pressure Curvature (PPC) Law of Gravity by establishing precise relationships among the gravitational pressure field (P_g), its spatial gradient (∇P_g), and the gravitational field (g). We demonstrate that gravitational

acceleration arises from the density-normalized gradient of the pressure field, thereby linking scalar pressure, vector force density, and observable acceleration into a single consistent framework. The formulation is shown to recover Newtonian gravity and Keplerian dynamics in the weak-field limit.

1. Introduction

In classical physics, gravity is described by the gravitational field:

$$\mathbf{g} = -\nabla \phi$$

where ϕ is the gravitational potential. In the PPC framework, gravity is interpreted as a pressure-driven phenomenon characterized by a scalar pressure field P_g .

This work establishes a direct connection between:

- Gravitational pressure field (P_g)
 - Pressure gradient field (∇P_g)
 - Gravitational field ($\mathbf{g}$)
-

2. Fundamental PPC Relation

The PPC law is expressed in force-density form:

$$\rho \mathbf{a} = -\nabla P_g$$

Since gravitational field is acceleration per unit mass:

$$\mathbf{g} = \mathbf{a}$$

we obtain:

$$\mathbf{g} = - (1/\rho) \nabla P_g$$

3. Relation to Gravitational Potential

In classical form:

$$\mathbf{g} = -\nabla \phi$$

Comparing with PPC expression:

$$-\nabla \varphi = - (1/\rho) \nabla P_g$$

This implies:

$$\varphi = P_g / \rho$$

Thus, gravitational potential is identified as pressure per unit density.

4. Unified Field Relations

The three fundamental quantities are related as follows:

1. Pressure field: P_g
2. Pressure gradient field: $\nabla P_g = -\rho g$
3. Gravitational field: $g = - (1/\rho) \nabla P_g$

These relations form a closed and consistent system.

5. Physical Interpretation

- P_g represents the scalar distribution of gravitational pressure in spacetime
- ∇P_g represents force density acting on matter
- g represents observable acceleration

Thus:

Pressure field $\rightarrow$ Gradient $\rightarrow$ Force density $\rightarrow$ Acceleration

6. Special Cases

6.1 Constant Pressure Field

If $P_g = \text{constant}$:

$$\nabla P_g = 0 \Rightarrow g = 0$$

No gravitational effects occur.

6.2 Strong Gradient Field

If ∇P_g is large:

g is large

Strong gravitational effects arise.

7. Weak-Field Limit

For a point mass M :

$$\phi = -GM/r$$

Using $\phi = P_g/\rho$:

$$P_g = -\rho GM/r$$

Taking gradient:

$$g = GM/r^2$$

Thus, the PPC formulation reproduces Newtonian gravity.

8. Circular Motion

For circular motion:

$$a = v^2 / r$$

Using PPC relation:

$$\rho v^2 / r = -\nabla P_g$$

This shows that centripetal force density is generated by pressure gradients.

9. Keplerian Consistency

From the above:

$$v^2 = GM/r$$

$$T^2 \propto r^3$$

Thus, Kepler's laws follow naturally from pressure-based dynamics.

10. Conclusion

The PPC framework establishes a direct and consistent connection between gravitational pressure field, its gradient, and the gravitational field. Gravity is interpreted as a manifestation of pressure gradients rather than a fundamental force. This unified formulation reproduces known physical laws while providing a deeper physical interpretation based on pressure.

Keywords

PPC Gravity, Gravitational Pressure, Pressure Gradient, Gravitational Field, Potential, Force Density, Kepler's Laws

Important Point:

Kepler's laws emerge when gravity follows an inverse-square law, which occurs in the weak-field limit of both Newtonian gravity and General Relativity.

In other words

Kepler's laws are valid in the weak-field (Newtonian) limit.

In the weak-field limit, where spacetime curvature is small and gravitational interactions follow an inverse-square law, Kepler's laws of planetary motion are recovered. Within the PPC framework, this corresponds to a regime in which the gravitational pressure gradient behaves as $\nabla P_g \propto 1/r^2$, leading to classical orbital dynamics.

The relation $F = \rho g$ is valid as a force-density expression in the weak-field (Newtonian) limit, where gravitational acceleration is well defined and spacetime curvature is negligible. In the PPC framework, this relation is consistent with $F = -\nabla P_g$, linking gravitational field to pressure gradients. However, in strong-field regimes, a covariant formulation is required.

Near the Neutron Star, the Gravitational Pressure Field is very large and the Pressure gradient is very large too.

- $P_g \rightarrow$ measures **intensity of gravitational pressure field**
- $\nabla P_g \rightarrow$ drives **dynamics (force / curvature effects)**

So:

👉 High P_g + high $\nabla P_g \Rightarrow$ **strong gravity regime**

Time Dilation and Its Validity

1. Time Dilation Formula (Special Relativity)

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Definitions:

- t_0 : Proper time (time measured in the rest frame)
- t : Observed time (moving frame)
- v : Relative velocity
- c : Speed of light

2. Physical Meaning

This equation describes **time dilation due to motion**:

- A moving clock runs **slower** compared to a stationary observer
 - As $v \rightarrow c$, time dilation becomes very large
-

3. Domain of Validity

This formula is valid when:

- Spacetime is **flat**
- **Gravitational effects are negligible**

👉 Therefore, it is **not specifically a weak-field formula**, but a **no-gravity (Special Relativity) result**

4. Weak Gravitational Field Case

When gravity is present but weak, time dilation includes both effects:

$$t \approx t_0 \left(1 + \frac{\phi}{c^2} + \frac{v^2}{2c^2} \right)$$

Where:

- ϕ = gravitational potential

👉 Time dilation arises from:

- Velocity (Special Relativity)
- Gravity (General Relativity)

The logic is:

Einstein Field Equation $\rightarrow g_{\mu\nu} \rightarrow g_{00} \rightarrow$ time dilation

In weak field:

$$g_{00} \approx -\left(1 + \frac{2\phi}{c^2}\right)$$

👉 From this, the time equation is derived.

5. Strong-Field Regime (Neutron Stars, Black Holes)

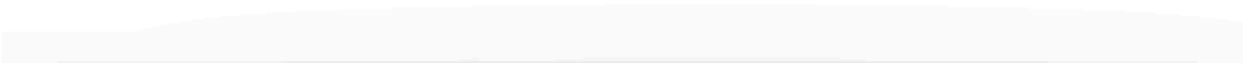
In strong gravity:

- Spacetime is **highly curved**
- The above formula is **not sufficient**

Time must be described using:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

👉 Time dilation depends on the **metric tensor**



6. Summary

- Special Relativity → time dilation due to motion
- General Relativity → time dilation due to gravity
- Real systems → combination of both

7. Conclusion

The relation

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

In gravitational fields—especially strong ones—it must be replaced or extended by a **metric-based description**.

Key Insight

Time dilation is a result of both motion and spacetime curvature—not motion alone.

$$\phi = \frac{P_g}{\rho}$$

So:

$$t \approx t_0 \left(1 + \frac{P_g}{\rho c^2} \right)$$

👉 This gives a **pressure-based time dilation interpretation**

$$E_d = \rho c^2, \quad E_d(r) = \rho(r) c^2,$$

$$P_g = \omega(r) \rho(r) c^2$$

👉 Now both:

- $\rho(r)$ varies
- $\omega(r)$ may vary

Then:

$$t \approx t_0 \left(1 + \frac{P_g(r)}{\rho(r)c^2} \right)$$

✓ Now time dilation depends on **position**

Position-Dependent Gravitational Pressure and Time Dilation

1. Gravitational Pressure Relation

$$P_g = \omega(r) \rho(r) c^2$$

Key Features:

- $\rho(r)$ → mass density varies with position
 - $\omega(r)$ → equation-of-state parameter
 - c → speed of light
-

2. Important Clarification

$\omega \neq \text{constant}$

- ω depends on the **physical state of matter**
- It may vary with **position** r
- Different regions (core, outer layers, vacuum) can have different values

Typical values (local, not universal):

- $\omega = 0 \rightarrow$ matter (dust)
- $\omega = \frac{1}{3} \rightarrow$ radiation
- $\omega = -1 \rightarrow$ vacuum energy
- $\omega = 1 \rightarrow$ stiff matter

3. Time Dilation Relation (Weak Field)

$$t \approx t_0 \left(1 + \frac{P_g(r)}{\rho(r)c^2} \right)$$

Substituting:

$$t \approx t_0 (1 + \omega(r))$$

4. Physical Interpretation

- Time dilation is now **position-dependent**
- It varies with:
 - local density $\rho(r)$
 - local parameter $\omega(r)$

👉 This ensures that time is **not constant across space**

5. Important Limitation

Even though:

$$t \approx t_0(1 + \omega(r))$$

this is:

- a **local approximation**
- not a complete gravitational description

👉 True gravity depends on **spatial variation of the field**, not only local values

6. Key Insight

- $\omega(r)$ describes **local matter properties**
 - $P_g(r), \rho(r)$ describe **spatial structure**
 - Time dilation arises from **variation in space**
-

7. Conclusion

By allowing both ρ and ω to vary with position, the PPC framework produces **position-dependent time dilation**. This is more physically realistic than constant-parameter models and aligns better with gravitational behavior in real systems.

Final Insight

Time dilation depends on spatial variation of physical fields, and ω must be treated as position-dependent, not constant.

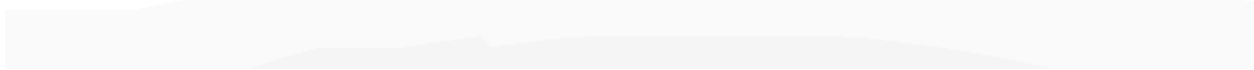
Position-Dependent Energy Density

$$E_d(r) = \rho(r)c^2$$

1. Meaning

- $E_d(r) \rightarrow$ **energy density as a function of position**
- $\rho(r) \rightarrow$ **mass density varies with radius r**
- $c \rightarrow$ speed of light

👉 This is the **local form** of mass–energy equivalence.

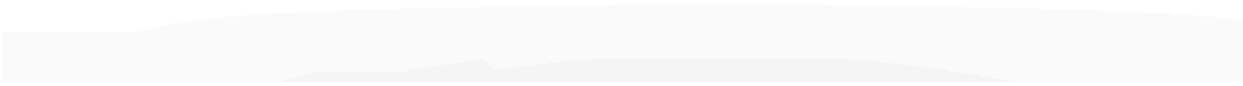


2. Physical significance

- Energy density is **not uniform in realistic systems**
- It changes with:
 - distance from the center
 - structure of the object (star, neutron star, etc.)

So:

👉 E_d must depend on **position**



$$P_g = \omega(r) \rho(r) c^2$$

Using:

$$E_d(r) = \rho(r) c^2$$

we get:

$$P_g(r) = \omega(r) E_d(r)$$

👉 This makes everything **position-dependent and consistent**



$$\rho(r) \rightarrow E_d(r) \rightarrow P_g(r) \rightarrow \nabla P_g(r)$$

👉 This leads to:

- spatial variation
- gravitational effects
- time dilation variation

1. If ω is not constant

Then it is a **scalar field**:

$$\omega = \omega(x^\mu) \quad \text{or (spherical symmetry)} \quad \omega = \omega(r)$$

So its gradient is **not zero**.

$$\nabla_{\mu}\omega = \partial_{\mu}\omega$$

- In curved spacetime, the covariant derivative of a scalar = ordinary derivative
- So:
 - Time component: $\partial_t\omega$
 - Spatial components: $\partial_i\omega$

If:

$$\omega = \omega(r)$$

then:

$$\nabla \omega = \frac{d\omega}{dr} \hat{r}$$

👉 The gradient points **radially**

$$\nabla\omega \neq 0$$

means:

- The **state of matter changes with position**
- Different regions have different physics:
 - Core vs outer layer
 - Dense vs less dense region

$$P_g = \omega(r) \rho(r) c^2$$

Take gradient:

$$\nabla P_g = c^2 [\rho(r) \nabla \omega(r) + \omega(r) \nabla \rho(r)]$$

👉 This is a **key result**

1. Density variation:

$$\omega \nabla \rho$$

2. Equation-of-state variation:

$$\rho \nabla \omega$$

- If $\omega = \text{constant}$:

$$\nabla\omega = 0$$

- If $\rho = \text{constant}$:

$$\nabla P_g \propto \nabla\omega$$

When ω is treated as a position-dependent scalar field, its gradient $\nabla_\mu \omega$ represents spatial variation in the local equation of state. In the PPC framework, this contributes directly to the pressure gradient through

$\nabla P_g = c^2[\rho \nabla \omega + \omega \nabla \rho]$, indicating that gravitational effects arise from both density variations and variations in the equation-of-state parameter.
